

Lecture 13

Theory of General Linear System III: Solving $Ax=b$

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Part 0 Review and Outline

3 mins

Recall

In the last lecture ...

- Definition of Linear independence
- Definition of Null space
- Definition of Span (of a collection of linear space elements)

Linear Dependence and Independence

Definition 12.1 (linear dependence)

Suppose V is a linear space over $\mathbb{R}$. Suppose $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in V$.

We say $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly dependent if

$$\exists \mathbf{c} = (c_1, c_2, \dots, c_k) \in \mathbb{R}^k, \mathbf{c} \neq \mathbf{0}, \text{ such that } c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k = \mathbf{0}.$$

Think: Can we remove condition $(c_1, \dots, c_k) \neq (0, 0, \dots, 0)$?

Definition 12.2 (linear independence)

Suppose V is a linear space over $\mathbb{R}$. Suppose $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in V$.

We say $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly independent

if $\forall \mathbf{c} = (c_1, c_2, \dots, c_k) \in \mathbb{R}^k \setminus \{\mathbf{0}\}$, we have

$$c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k \neq \mathbf{0}.$$

Or: $c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k = \mathbf{0}$ only happens when $c_1 = \dots = c_k = 0$.

Remark: We will say the set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is linearly independent (or dependent) if the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly independent (or dependent).

Homogeneous Linear System

Definition (homogeneous linear system)

A homogeneous linear system is

$$Ax = 0$$

where $A \in \mathbb{R}^{m \times n}$ are given and $x \in \mathbb{R}^{n \times 1}$ is the variable

Theorem 12.1

The solution set of a homogeneous linear system $Ax=0$ is a linear space.

Namely: $\{x \in \mathbb{R}^n | Ax = 0\}$ is a linear space.

Definition 12.3 (null space)

The solution set of $Ax=0$ is called the null space of A , denoted as $N(A)$.

Recall: Span

Definition 12.4 (span)

Suppose V is a linear space.

Suppose $\mathcal{U} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is a subset of V .

The span of $\mathcal{U}$ is defined as $\text{span}(\mathcal{U}) \triangleq \{a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k \mid a_1, \dots, a_k \in \mathbb{R}\}$,

In words: the span (of elements of a linear space)
is the set of all linear combinations of these elements.

Eg: $W = \{s\mathbf{u} + t\mathbf{v} \mid s, t \in \mathbb{R}\}$ is the span of $\{\mathbf{u}, \mathbf{v}\}$.

Remark: For simplicity, we can also say W is the span of $\mathbf{u}, \mathbf{v}$.

Equivalent statements:

$\{a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k \mid a_1, \dots, a_k \in \mathbb{R}\}$ is the span of $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$.

$\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ spans $\{a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k \mid a_1, \dots, a_k \in \mathbb{R}\}$

$\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is a spanning set of $\{a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k \mid a_1, \dots, a_k \in \mathbb{R}\}$

Lec 13 Solving $Ax=b$ & Linear Independence

Outline:

1. Column space
2. Computing the solution set of $Ax = b$
3. Basis (time permits)

Strang's book v5:
Sec 3.2, 3.3, 3.4

Learning goals: After this lecture, you should be able to

1. Express the column space of a matrix and evaluate if a vector belongs to this column space or not
2. Write the solution set of $Ax=b$ via language of linear space
3. Find basis of common linear spaces (time permits)

Part I: Column space

Strang's book: Sec 3.3

Definition: Column Space

Definition (column space)

Suppose $A = [\mathbf{a}_1, \dots, \mathbf{a}_n] \in \mathbb{R}^{m \times n}$ is a matrix.

Then $\text{span}(\{\mathbf{a}_1, \dots, \mathbf{a}_n\})$ is called the column space of A , denoted as $C(A)$.

**In words: A 's column space is the span of A 's column vectors;
i.e., the set of all linear combinations of columns.**

Eg: $C(I_n) = \underline{\mathbb{R}^n}$

Eg: Column space of $A = \begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 3 \end{bmatrix}$ is the set $\left\{ \alpha_1 \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix} \mid \alpha_1, \alpha_2 \in \mathbb{R} \right\}$.

Have you seen this before?

Column Space and Linear Space

Eg: Column space of $A = \begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 3 \end{bmatrix}$ is the set $\left\{ \alpha_1 \underbrace{\begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}}_{\vec{a}_1} + \alpha_2 \underbrace{\begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix}}_{\vec{a}_2} \mid \alpha_1, \alpha_2 \in \mathbb{R} \right\} = C(A)$ [column form]

or set $\left\{ \begin{bmatrix} \alpha_1 \\ 4\alpha_1 + 3\alpha_2 \\ 2\alpha_1 + 3\alpha_2 \end{bmatrix} \mid \alpha_1, \alpha_2 \in \mathbb{R} \right\}$ [scalar form], or set $\{ A\vec{x} \mid \vec{x} \in \mathbb{R}^2 \} = C(A)$ [matrix form]

Let's see what "a vector lies in column space" means.

Column form:

$\underline{\mathbf{b}} \in C(A) \iff \exists \alpha_1, \alpha_2, \text{ s.t. } \alpha_1 \vec{a}_1 + \alpha_2 \vec{a}_2 = \mathbf{b} \iff \chi_1 \vec{a}_1 + \chi_2 \vec{a}_2 = \mathbf{b}$ has at least one solution $\mathbf{x}$

Matrix form:

$\mathbf{b} \in C(A) \iff \exists \alpha_1, \alpha_2, \text{ s.t. } A\vec{x} = \mathbf{b} \iff A\mathbf{x} = \mathbf{b}$ has at least one solution $\mathbf{x}$

Property: $A \in \mathbb{R}^{m \times n}$. Then $C(A) = \{ A\vec{x} \mid \vec{x} \in \mathbb{R}^{n \times 1} \}$.

Remarks

Column space is “attached to” any matrix A .

Column space of A is defined JUST based on A .

Just like: rows of A , *inverse of A* are defined based on A .

Column space of A is **NOT** defined based on linear system $Ax=b$.

But....

there is a relation...

Column space and solvability

Proposition 12.1

$Ax=b$ has a solution iff $b \in C(A)$.

Remark: The first result in this class about **solvability** of linear system.
—more will come later.

Proof (Method 1): $A\vec{x} = \vec{b}$ has a solution $A = [\vec{a}_1, \dots, \vec{a}_n]$
(column form)

$$\begin{aligned} &\iff \exists \vec{x} \in \mathbb{R}^n \text{ such that } A\vec{x} = \vec{b} \\ &\iff \exists \vec{x} \in \mathbb{R}^n \text{ such that } x_1 \vec{a}_1 + \dots + x_n \vec{a}_n = \vec{b} \\ &\iff \vec{b} \in \text{span} \{ \vec{a}_1, \dots, \vec{a}_n \} \\ &\iff \vec{b} \in C(A) \end{aligned}$$

Column space and solvability

Proposition 12.1

$Ax=b$ has a solution iff $b \in C(A)$.

Remark: The first result in this class about **solvability** of linear system.
—more will come later.

Proof (Method 2) : $A\vec{x} = \vec{b}$ has a solution
(matrix form)

$$\iff \exists \vec{x} \in \mathbb{R}^n \text{ such that } A\vec{x} = \vec{b}$$

$$\iff \vec{b} \in \{A\vec{x} \mid \vec{x} \in \mathbb{R}^n\}$$

$$\iff \vec{b} \in C(A)$$

Existence of solution & $C(A)$

Ex1: $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ does $Ax = b$ has a solution?

Yes because

$$b \in \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\} \\ = C(A).$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

Ex2: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ does $Ax = b$ has a solution?

No

because

$$b \notin \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\} = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\} = \left\{ t \begin{bmatrix} 1 \\ 2 \end{bmatrix}, t \in \mathbb{R} \right\}$$

$\parallel$
 $2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$t \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \forall t.$$

Part II Solution Set of $Ax=0$ and $Ax=b$

Strang's book: Sec 3.2, Sec. 3.3

- Expressing null space by span
- Matrix expressions

Null Space & Span

We learned null space and span.

Next: relation of null space and span.

Solving a Homogeneous Linear System

We will study the solution set of $Ax=0$, i.e., a null space.

Use the following system as an example.

We will solve the system, to obtain its solution set.

$$\begin{cases} x_1 + x_2 + 2x_4 + 5x_5 = 0, \\ x_2 + 3x_4 + x_5 = 0, \\ x_1 + x_3 + x_4 + 6x_5 = 0, \\ x_1 + 2x_2 + x_3 + 4x_4 + 7x_5 = 0. \end{cases} \quad \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 2 & 5 & 0 \\ 0 & 1 & 0 & 3 & 1 & 0 \\ 1 & 0 & 1 & 1 & 6 & 0 \\ 1 & 2 & 1 & 4 & 7 & 0 \end{array} \right]$$

Revisit: Infinitely Many Solutions for Homogeneous System

Original system

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 2 & 5 & 0 \\ 0 & 1 & 0 & 3 & 1 & 0 \\ 1 & 0 & 1 & 1 & 6 & 0 \\ 1 & 2 & 1 & 4 & 7 & 0 \end{array} \right]$$

GJ-E

RREF

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 4 & 0 \\ 0 & 1 & 0 & 3 & 1 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Equivalent linear system:

$$\begin{cases} x_1 - x_4 + 4x_5 = 0, \\ x_2 + 3x_4 + x_5 = 0, \\ x_3 + 2x_4 + 2x_5 = 0, \\ 0 = 0. \end{cases} \Rightarrow \begin{cases} x_1 = \underline{s} - 4\underline{t}, \\ x_2 = -3\underline{s} - \underline{t}, \\ x_3 = -2\underline{s} - 2\underline{t}, \\ 0 = 0. \end{cases}$$

Solution set:

$$W = \{ [s - 4t, -3s - t, -2s - 2t, s, t]^T \mid s, t \in \mathbb{R} \} \subseteq \mathbb{R}^{5 \times 1}$$

Thm 11.1 says: $W = \text{Null}(A)$ must be a linear space.

Solution Set Written As a Span

Solution set of $Ax=0$:

$$= \{s\vec{u} + t\vec{v} \mid s, t \in \mathbb{R}\}$$

$$W = \{[s - 4t, -3s - t, -2s - 2t, s, t]^T \mid s, t \in \mathbb{R}\} \subseteq \mathbb{R}^{5 \times 1}$$

This must be a linear space (since $\{x: Ax=0\}$ is a linear space).

Can we directly show this is a linear space, based on this form?

$$\text{Note that } \begin{bmatrix} \underline{s} - \underline{4t} \\ \underline{-3s} - \underline{t} \\ \underline{-2s} - \underline{2t} \\ \underline{s} \\ \underline{t} \end{bmatrix} = s \cdot \underbrace{\begin{bmatrix} 1 \\ -3 \\ -2 \\ 1 \\ 0 \end{bmatrix}}_{\vec{u}} + t \underbrace{\begin{bmatrix} -4 \\ -1 \\ -2 \\ 0 \\ 1 \end{bmatrix}}_{\vec{v}} = s\vec{u} + t\vec{v}$$

So $W = \text{span} \{ \vec{u}, \vec{v} \}$,

We can say more than "Null space is a linear space":

Null space is a span of certain vectors.

Conclusion: Null Space Written As a Span

$$\begin{cases} x_1 + x_2 + 2x_4 + 5x_5 = 0, \\ x_2 + 3x_4 + x_5 = 0, \\ x_1 + x_3 + x_4 + 6x_5 = 0, \\ x_1 + 2x_2 + x_3 + 4x_4 + 7x_5 = 0. \end{cases}$$

↓ solve

$$W = \{[s - 4t, -3s - t, -2s - 2t, s, t]^T \mid s, t \in \mathbb{R}\} \subseteq \mathbb{R}^{5 \times 1}$$

↓ rewrite

$$\begin{aligned} W &= \{s\mathbf{u} + t\mathbf{v} \mid s, t \in \mathbb{R}\} \\ &= \text{span}\{\mathbf{u}, \mathbf{v}\} \end{aligned}$$

where $\mathbf{u} = \begin{bmatrix} 1 \\ -3 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -4 \\ -1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$

Homogeneous linear system

$$A\mathbf{x} = \mathbf{0}$$



Solution set

$$W = \text{span}\{u_1, \dots, u_p\}.$$

Solution set of $Ax = b$?

Previous pages:

We know how to write the solution set of $Ax=0$ in terms of linear space.

Next, we consider $Ax = b$.

It only requires minor extra work.

Consider the following system as an example.

$$\begin{cases} x_1 + x_2 + 2x_4 + 5x_5 = 7, \\ x_2 + 3x_4 + x_5 = 6, \\ x_1 + x_3 + x_4 + 6x_5 = 8, \\ x_1 + 2x_2 + x_3 + 7x_4 + 8x_5 = 20 + c. \end{cases}$$

Same LHS as prev. example.
nonzero RHS.

Example of General System

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 2 & 5 & 7 \\ 0 & 1 & 0 & 3 & 1 & 6 \\ 1 & 0 & 1 & 1 & 6 & 8 \\ 1 & 2 & 1 & 7 & 8 & 20+c \end{array} \right] \xrightarrow{\text{GJ-E}} \left[\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 4 & 1 \\ 0 & 1 & 0 & 3 & 1 & 6 \\ 0 & 0 & 1 & 2 & 2 & 7 \\ 0 & 0 & 0 & 0 & 0 & c \end{array} \right]$$

Equivalent linear system:

$$\begin{cases} x_1 - x_4 + 4x_5 = 1, \\ x_2 + 3x_4 + x_5 = 6, \\ x_3 + 2x_4 + 2x_5 = 7, \\ 0 = c. \end{cases} \Rightarrow \begin{cases} x_1 = \overset{s}{x_4} - 4\overset{t}{x_5} + 1, \\ x_2 = -3x_4 - x_5 + 6, \\ x_3 = -2x_4 - 2x_5 + 7, \\ 0 = c. \end{cases}$$

Case 1: $c \neq 0$. Solution set is empty.

one particular solution:

pick $x_4 = x_5 = 0$,
then $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \\ 7 \end{pmatrix}$.

Case 2: $c = 0$. The solution set is

$$\left\{ (s - 4t + 1, -3s - t + 6, -2s - 2t + 7, s, t) \mid s, t \in \mathbb{R} \right\}.$$

Is it a linear space?

Example of General System: Expression By 3 Vectors

$$W = \{[\alpha - 4\beta + 1, -3\alpha - \beta + 6, -2\alpha - 2\beta + 7, \alpha, \beta]^T \mid \alpha, \beta \in \mathbb{R}\} \subseteq \mathbb{R}^{5 \times 1}$$

is NOT a linear space. *not span of $\vec{u}, \vec{v}$.*

Each solution can be written as

of $Ax=b$

$$x = \underbrace{\begin{bmatrix} 1 \\ 6 \\ 7 \\ 0 \\ 0 \end{bmatrix}}_{\vec{x}_p} + \alpha \underbrace{\begin{bmatrix} 1 \\ -3 \\ -2 \\ 1 \\ 0 \end{bmatrix}}_{\vec{u}} + \beta \underbrace{\begin{bmatrix} -4 \\ -1 \\ -2 \\ 0 \\ 1 \end{bmatrix}}_{\vec{v}}, \forall \alpha, \beta \in \mathbb{R}.$$

$\vec{x}_p + (\text{LC of } \vec{u}, \vec{v})$.

What can we say about the 3 vectors?

$\vec{u}, \vec{v}$ are solutions of $A\vec{x} = 0$.

$\vec{x}_p$ is a solution of $A\vec{x} = b$.

pick $\alpha = \beta = 0$, $x = x_p$

Solution of $Ax=b$ and Solution of $Ax=0$

The following result holds not only for the example, but general system.

Claim: If $\mathbf{x}_p$ is a solution of $Ax=b$, then any solution of $Ax=b$ can be written as $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$, where $\mathbf{x}_n \in N(A)$.

General Proof:

Suppose $\mathbf{x}$ is a solution of $A\mathbf{x} = b$,

Then $A\mathbf{x} = b$ (1)

We also have $A\mathbf{x}_p = b$ (2).

Denote $\mathbf{x}_n = \mathbf{x} - \mathbf{x}_p$.

Then $\mathbf{0} = b - b = (A\mathbf{x} - A\mathbf{x}_p) = A(\vec{x} - \vec{x}_p) = A\vec{x}_n$.

Thus $\mathbf{x}_n \in N(A)$.

Diff of two solutions
of $AX=b$ is a solution
of $AX=0$.

Notation: Element + Set

Any solution of $Ax=b$ can be written as $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$,
where $\mathbf{x}_n \in N(A)$.

We introduce a special operation: element plus set.

Definition: For any element $\mathbf{v}$ and a set U , define

$$\mathbf{v} + U \triangleq \{\mathbf{v} + \mathbf{u} \mid \mathbf{u} \in U\}$$

Eg1: ^{number} $\textcircled{-1} + \textcircled{\text{set } \{3,5\}} = \{\text{set } \boxed{-1+3}, \boxed{-1+5}\} = \{\text{set } \boxed{2,4}\}.$

Eg2: $1 + \underbrace{\{\text{even numbers}\}}_{\text{set}} = \{\underbrace{\text{odd numbers}}_{\text{set}}\}.$

Theorem on Solution Set of $Ax=b$

If $\mathbf{x}_p$ is a solution of $Ax=b$, then any solution of $Ax=b$ can be written as $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$.

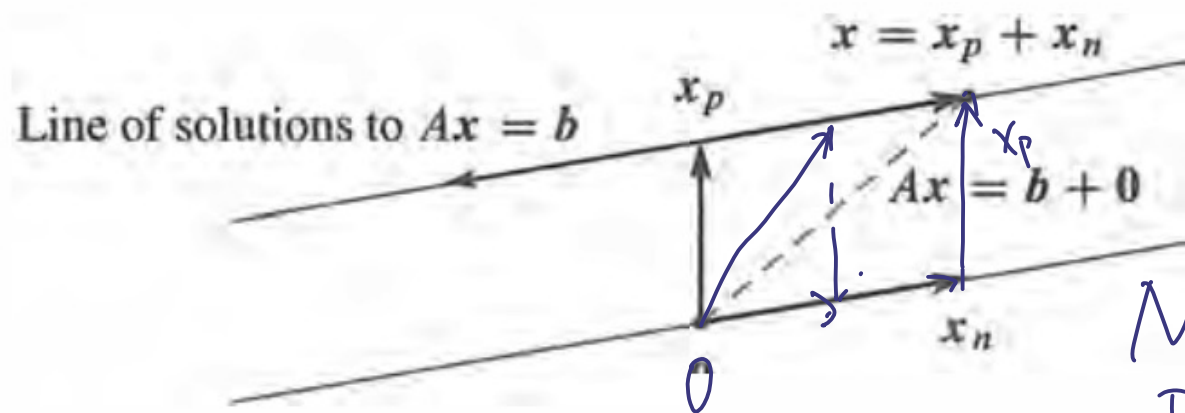
With the notation defined earlier, we have:

Proposition 13.1

The solution set of a linear system $A\mathbf{x} = \mathbf{b}$ is:
either (i) an empty set

or (ii) $\mathbf{x}_p + N(A)$, where $\mathbf{x}_p$ is any solution of $A\mathbf{x} = \mathbf{b}$.

$\mathbf{x}_p + N(A)$: informally, a “shifted” linear space



$x_p + N(A)$: not a linear space
does NOT pass "0"

$N(A)$: linear space
pass 0

("same" shape:
flat.)

Complete solution = one particular solution + all nullspace solutions.

Exercise: True or False

The solution set of a linear system can

1. have exactly 3 elements.

F

2. be $\{0, \pm 2, \pm 4, \pm 6, \dots\}$.

F

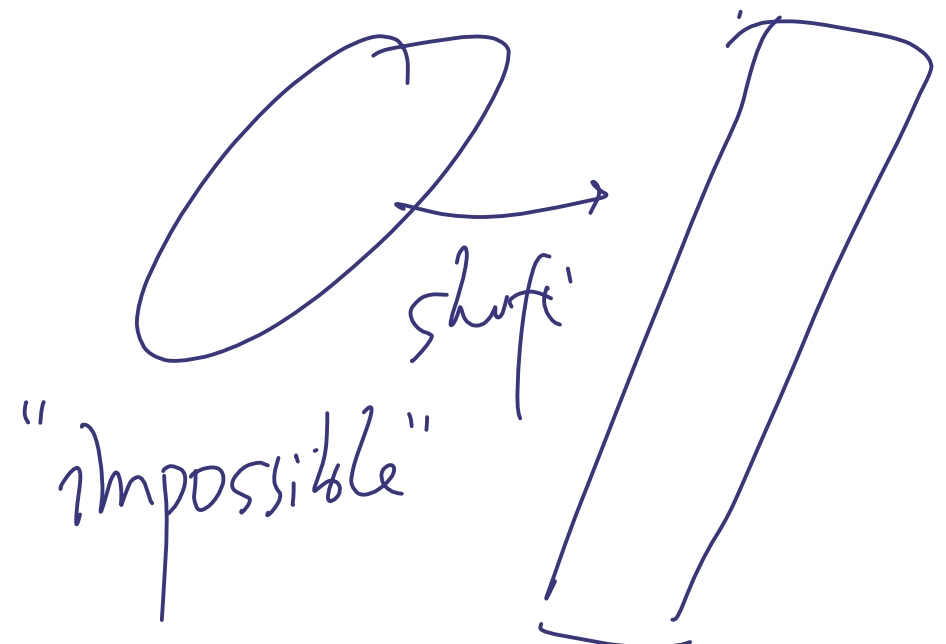
3. be an ellipse given by $\{(x_1, x_2) \mid \frac{x_1^2}{4} + \frac{x_2^2}{9} = 1\}$.

F

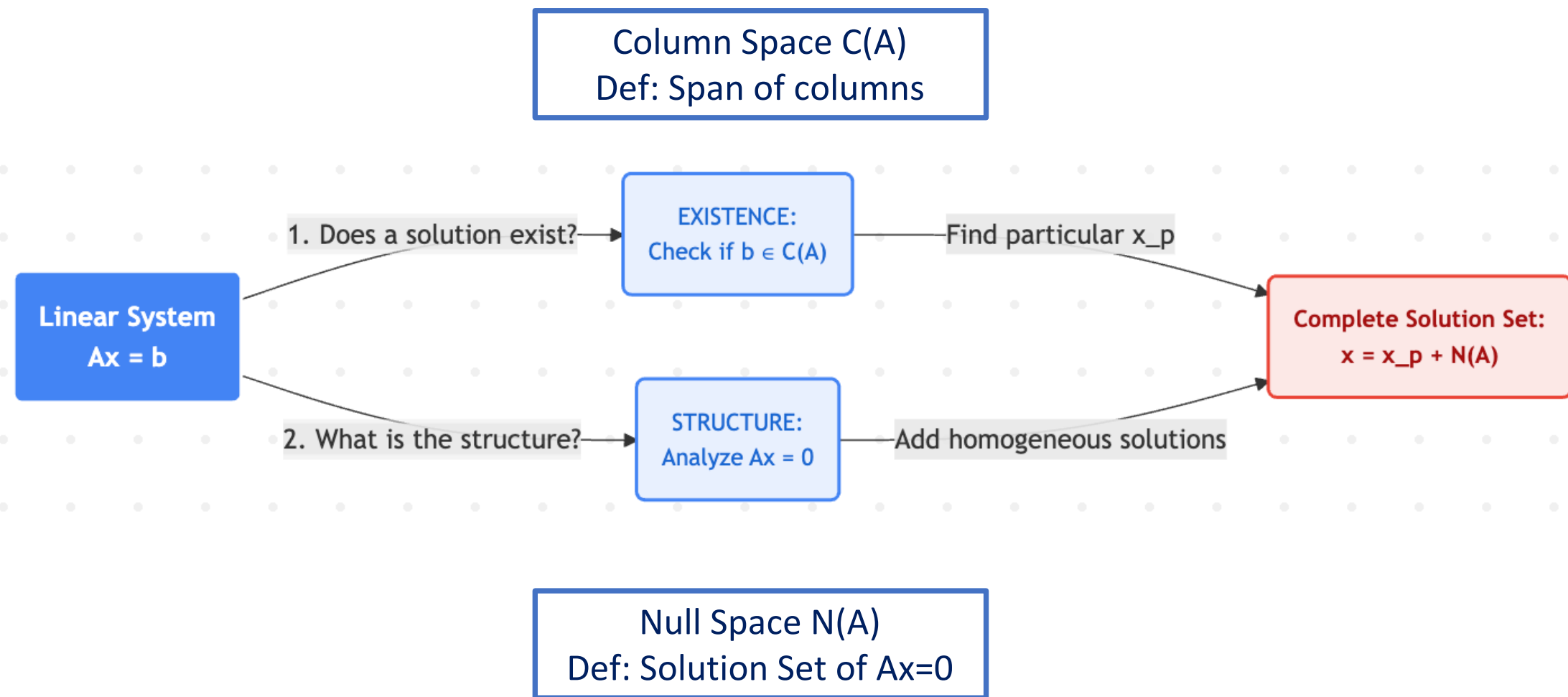
discrete

-4 -2 0 2 4
• • • • •

cannot be obtained by
shifting a linear space



Summary So Far



Reading: Relation to Inverse

For “good” square linear system $A\mathbf{x} = \mathbf{b}$, the solution is $A^{-1}\mathbf{b}$.

For general rectangular system $A\mathbf{x} = \mathbf{b}$, the solution is $\mathbf{x}_p + N(A)$.

How are they related?

Actually, $\mathbf{x}_p = B^{-1}\mathbf{b}_p$

for certain matrix B obtained from A ,
and certain vector $\mathbf{b}_p$ obtained from $\mathbf{b}$.

Skip the derivation here.

Part III Definition of Basis

Strang's book Sec. 3.4
Part "A Basis for a Vector Space"

Basis



Definition 13.1 (basis)

Suppose V is a linear space over $\mathbb{R}$.

Suppose $\mathcal{U} \triangleq \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\} \subseteq V$. We say $\mathcal{U}$ is a basis if

- (i) $\mathcal{U}$ is linearly independent;
- (ii) $\text{span}(\mathcal{U}) = V$.

In words, a basis of V is linearly independent & spanning set of V .

Eg1 $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a basis of $\mathbb{R}^n$. (called “standard basis”)

Eg2 $\{\mathbf{u}\}$ is a basis of $\text{span}\{\mathbf{u}\}$, if $\mathbf{u} \neq \mathbf{0}$.

[explan (etc)]

Checking Basis of V

Problem: Find a basis of a linear space V .

Preparation: Find a candidate set $\mathcal{U}$.

First (independence)

Check the elements of $\mathcal{U}$ are linearly independent.

Second (span)

Check any element of V can be expressed as a linear combination of elements of $\mathcal{U}$.

$$V \stackrel{?}{=} \text{span}(\mathcal{U}) = \{ \text{LC of } \mathcal{U} \}.$$

$\downarrow$
 $\mathcal{U} \stackrel{\text{want}}{=} \text{LC of } \mathcal{U}.$

Example: Show A Set is Basis

Prove: The set of vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ is a basis of $\mathbb{R}^3$.

Proof:

$\ell_1 \quad \ell_2 \quad \ell_3$

Spanning Set

For any vector $\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \in \mathbb{R}^3$, it can be expressed as $\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = v_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + v_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Thus the three vectors span $\mathbb{R}^3$.

Linear Independence

Suppose $\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.

Then, $\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ which implies $\alpha_1 = \alpha_2 = \alpha_3 = 0$.

Thus the three vectors are linearly independent.

Exercise: Show A Set is Basis

Prove: The set of vectors $\underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\ell_1}, \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\ell_2}, \underbrace{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}_{\ell_1 + \ell_2}$ is a basis of $\mathbb{R}^3$.

Proof:

i) Spanning set.

For any vector $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3$, it can be expressed as $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \boxed{(v_1 - v_3)} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \boxed{v_3} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

Thus the three vectors span $\mathbb{R}^3$.

$\cancel{v_1} = \begin{bmatrix} (v_1 - v_3) + v_3 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

ii) Linear independence.

Suppose $\alpha_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Then, the vector $\begin{bmatrix} \alpha_1 + \alpha_3 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, which implies $\alpha_1 + \alpha_3 = \alpha_2 = \alpha_3 = 0$.

$\alpha_1 = 0$

Thus, $\alpha_1 = \alpha_2 = \alpha_3 = 0$. Thus the three vectors are linearly independent.

Non-Example: Show A Set is Not a Basis

The set of vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ is NOT a basis of $\mathbb{R}^3$.

Since they cannot span $\mathbb{R}^3$ (cannot represent $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$)
 $\alpha \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

The set of vectors $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ is NOT a basis of $\mathbb{R}^3$.
 $\vec{e}_1, \vec{e}_2, \vec{e}_3, \vec{v}$

Since they are NOT independent.

$$\begin{aligned} \vec{u} &= \alpha \vec{e}_1 + \beta \vec{e}_2 + \gamma \vec{e}_3 + \delta \vec{v} \\ &= \alpha' \vec{e}_1 + \beta' \vec{e}_2 + \gamma' \vec{e}_3 \end{aligned}$$

Remark: $\vec{v} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ is a LC of $\underline{e_1, e_2, e_3}$, so it is redundant
for representing vectors in $\mathbb{R}^3$.

Basis of Matrix Space and Polynomial Space

— Find a basis of $\mathbb{R}^{2 \times 2}$.

$$\{E_{11}, E_{12}, E_{21}, E_{22}\} \quad \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

— Find a basis of $\underline{P_2} \triangleq \{f(x) = a_0 + a_1x + a_2x^2 : a_0, a_1, a_2 \in \mathbb{R}\}$

Define $f_0(x) = 1, \forall x \in \mathbb{R};$

$$f_1(x) = x, \forall x \in \mathbb{R};$$

$$f_2(x) = x^2, \forall x \in \mathbb{R}.$$

Then $\{f_0, f_1, f_2\}$ is a basis of P_2 .

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Polynomial: f_0 , or $\underbrace{f_0(\cdot)}_{\text{emphasize "function"}}$, or $\underbrace{f_0(x), f_0(t)}_{\text{convention, keep } x}$

$$f_1(x) = x, \forall x \in \mathbb{R}$$

same as

$$f_1(t) = t, \forall t \in \mathbb{R},$$

same as

$$f_1(z) = z, \forall z \in \mathbb{R}.$$

Symbol x, t, z
do not matter.

Basis of Matrix Space

The set B consisting of

$$\underline{B_{11}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, B_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, B_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

is a basis of $\mathbb{R}^{2 \times 2}$. B is linearly independent. Also notice that $\forall A \in \mathbb{R}^{2 \times 2}$,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

In general, $\{E_{ij}, i = 1, \dots, m; j = 1, \dots, n\}$ is a basis of $\mathbb{R}^{m \times n}$,
where E_{ij} has only (i, j) -th entry being 1 and other entries being 0.

$$E_{ij} = \begin{bmatrix} 0 & \cdots & \overset{j\text{-th col}}{1} & \cdots & 0 \\ 0 & \cdots & \vdots & \cdots & \vdots \\ \vdots & \cdots & 1 & \cdots & \vdots \\ 0 & \cdots & \vdots & \cdots & \vdots \end{bmatrix}$$

i-th row

only one "1"
in the matrix

Basis of Polynomial Space

Find a basis of the vector space P_2 , where P_2 is defined as

$$P_2 \triangleq \{f(x) = a_0 + a_1x + a_2x^2 : a_0, a_1, a_2 \in \mathbb{R}\}.$$

Define the polynomials $f_0(x) = 1$, $f_1(x) = x$, and $f_2(x) = x^2$.

Then the set $\{f_0, f_1, f_2\}$ is a basis of P_2 .

Proof: (1) Verify spanning set. Suppose $f(x) = a_0 + a_1x + a_2x^2$

$$f(x) = a_0 \cdot f_0(x) + a_1 f_1(x) + a_2 f_2(x), \text{ a linear combination of } f_0, f_1, f_2.$$

(2) Verify independence.

Assume $\alpha \cdot f_0(x) + \beta f_1(x) + \gamma f_2(x) = \mathbf{0}$ (the zero polynomial),

$$\text{This means } \alpha + \beta x + \gamma x^2 = 0, \forall x \in \mathbb{R},$$

which implies $\alpha = \beta = \gamma = 0$. (prove it next page)

$$\begin{aligned} & \Leftrightarrow g = 0 \text{ (zero polynomial)} \\ & g(x) = 0, \forall x \in \mathbb{R} \end{aligned}$$

Proof of an Intermediate Result

Claim If $\alpha + \beta x + \gamma x^2 = \underline{0}$ for all $x \in \mathbb{R}$, then $\alpha = \beta = \gamma = 0$.

Proof:

Pick $x = 0$: $\implies \underline{\alpha = 0}$.

Pick $x = 1$: $\implies \alpha + \beta + \gamma = 0$.

Pick $x = 2$: $\implies \alpha + 2\beta + 4\gamma = 0$.

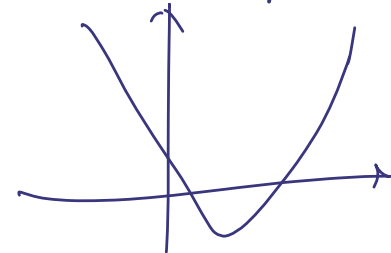
We obtain this linear system pick $x=3, 4, 5, \dots$

$$\begin{cases} \alpha = 0, \\ \alpha + \beta + \gamma = 0 \\ \alpha + 2\beta + 4\gamma = 0 \end{cases} \implies \begin{cases} \alpha = 0 \\ \beta = 0 \\ \gamma = 0 \end{cases}$$

Solving the system, we get $\underline{\alpha = \beta = \gamma = 0}$.

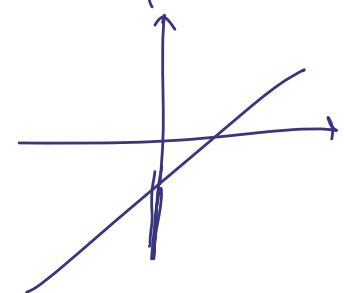
Geometry.

If $\gamma \neq 0$, quadratic function

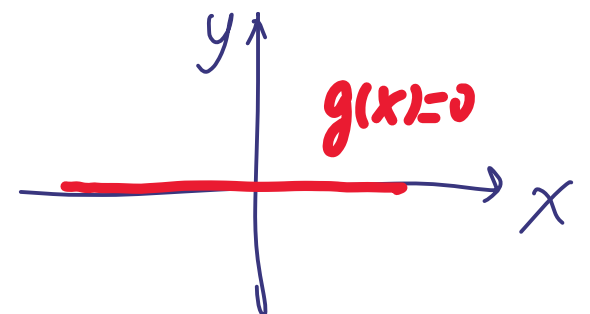


cannot be all zero.

If $\gamma = 0$, $\beta \neq 0$, linear function



zero polynomial



Judgement: True or False

1) If $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in V$ are linearly independent, then they form a basis of V .

F

2) If $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in V$ are linearly independent, then they form a basis of $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$.

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3) If $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ span V , then they form a basis of V .

F

Summary Today (write Your Own)

One sentence summary:

Quiz 2,
Lex 1-12.

Detailed summary:

Summary Today (of Instructor)

One sentence summary:

We study how to solve $Ax=b$; linear independence, basis and dimension.

Detailed summary:

1. Solving $Ax=b$

- Solution set of $Ax=0$: Null space $N(A)$
- Solution set of $Ax=b$: $\mathbf{x}_p + N(A)$ or empty set

2. Linear dependence

- Linear dependent elements: trivial linear combination gets 0
- Related to $Ax=0$ having infinitely many solutions

3. Basis

- Basis of V : independent & spanning set of V
- Bases of Euclidean space, Matrix Space, Polynomial Space

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